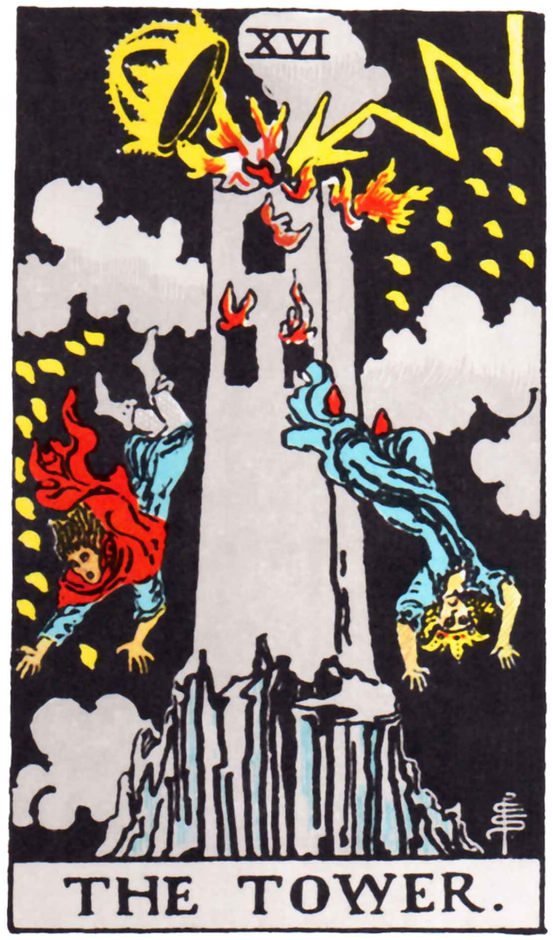
The Byzantine Generalization Problem

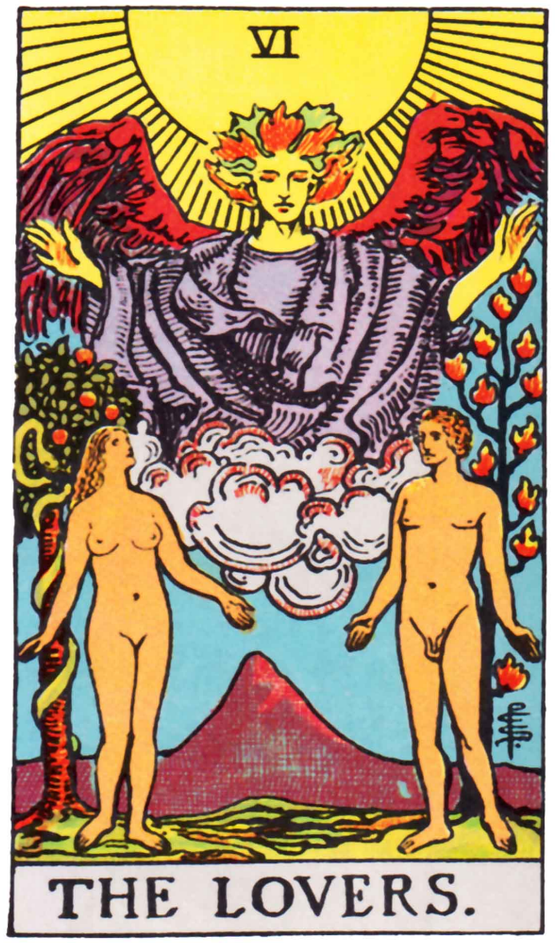
Governance and its Malcontents

Tragedy of the Commons



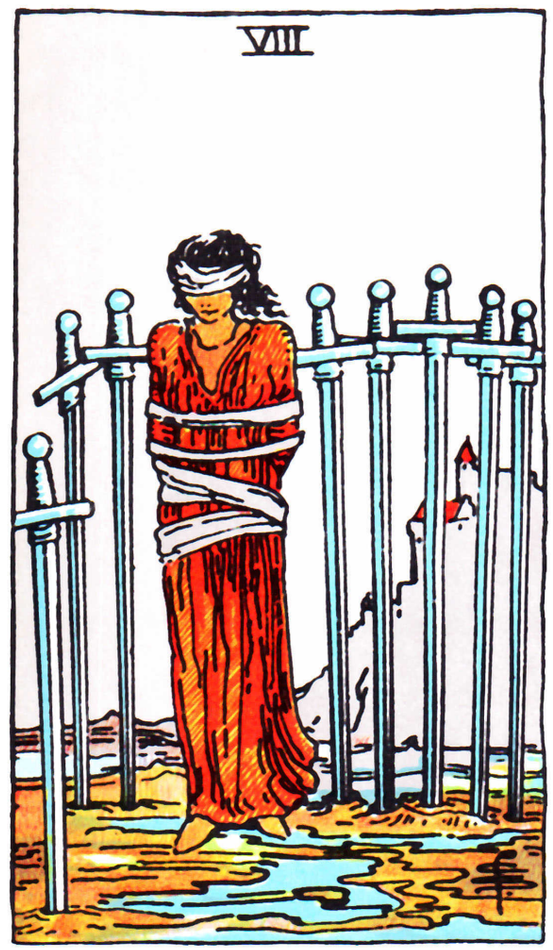
The tragedy of the commons describe a situation in a shared-resource system where individual users acting independently according to their own self-interest behave contrary to the common good of all users by depleting or spoiling that resource through their collective action.

Prisoner’s Dilemma



The prisoner's dilemma describes a scenario in which both parties choose to protect themselves at the expense of the other participant.

Zero-Sum



Zero-sum is a situation in which each participant's gain or loss of utility is exactly balanced by the losses or gains of the utility of the other participants. If the total gains of the participants are added up and the total losses are subtracted, they will sum to zero.

The Byzantine Generals' Problem

It is an agreement problem, in which a group of commanding generals encircle a city and formulate a plan for attacking the city. Every general should agree on a common decision—attack or retreat—and accurately communicate it to the other generals.

Governance

Governance is “the process of decision-making and the process by which decisions are implemented (or not implemented).”

Vlad Zamfir
@VladZamfir

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Updated definition of governance:

The process of establishing, maintaining and revoking the legitimacy of decisions and decision making processes, norms, and principles

How do Developers, Miners, Nodes, Users, and Coin and Token Holders make decisions about the blockchain outside of the current technical specifications and incentives?

On-Chain Governance

That is, something like incorporating weighted voting systems for stakeholders.

On-Chain Governance

1 “Person” = 1 Vote
Problem: You can create infinite accounts

1 Token = 1 Vote
Problem: You get plutarchy

1 Verified Identity = 1 Vote
1 50% Verified Identity = 0.5 Vote
1 Anonymous “Identity” = 0.25 Vote
Question: Do we want verified blockchain identities?

Off-Chain Governance

Governance is a complex social process, that shouldn't fully be encoded, and proposals for such threaten the legitmacy of current stakeholders.

Nick Szabo ⚡
@NickSzabo4

Following

Applying the word "governance" to the residual human decision-making needed for public blockchains carries, as the term is thought of by non-developers, tons of baggage that is irrelevant to the only necessary category of such decisions, namely decisions about upgrading software.

6:44 PM - 13 Feb 2018

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Replying to @NickSzabo4

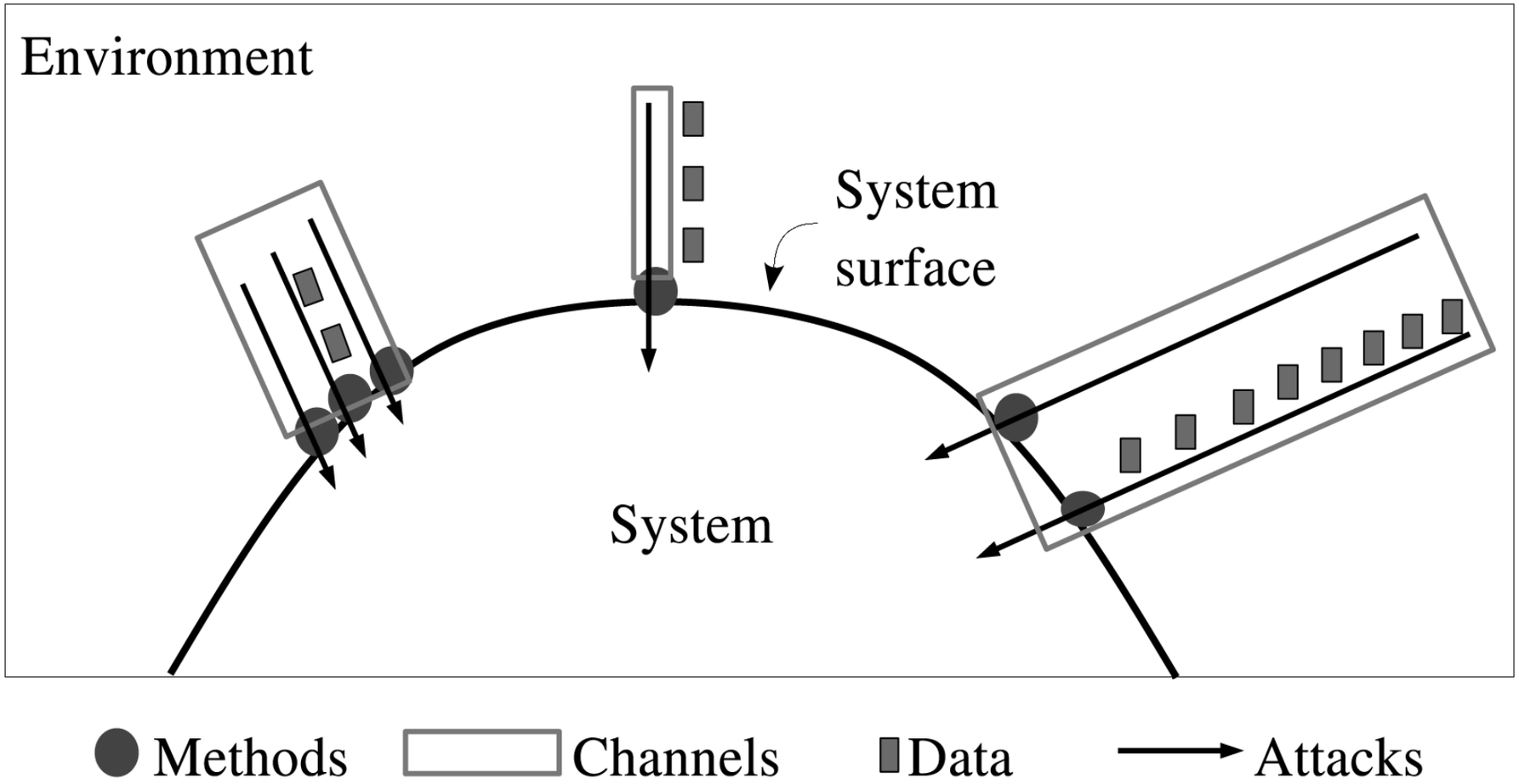
What do you think the proper definition/use of the term "governance" for public blockchains? Because apparently "governance" does not mean compliance with decisions backed by enforcement.

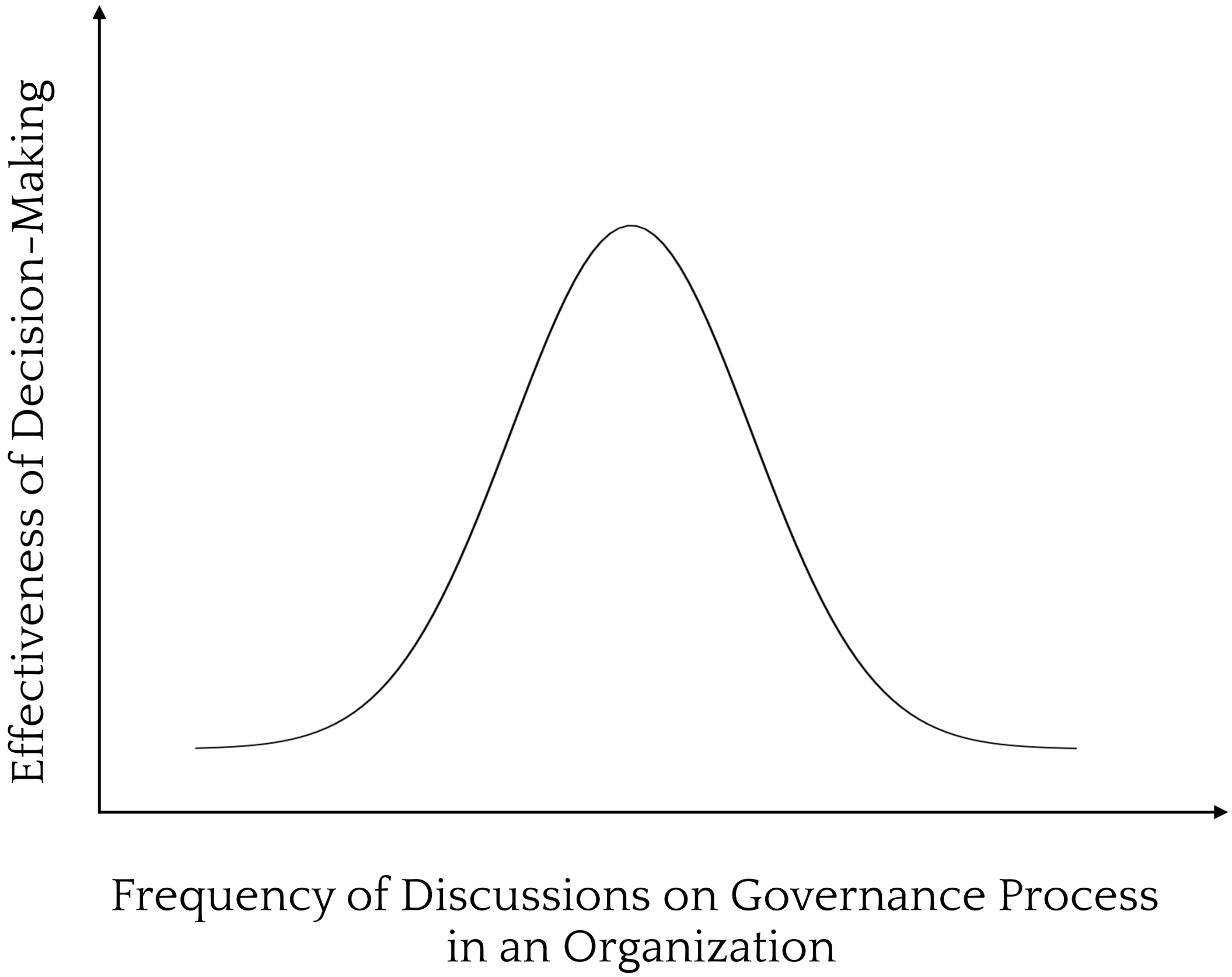
2 4

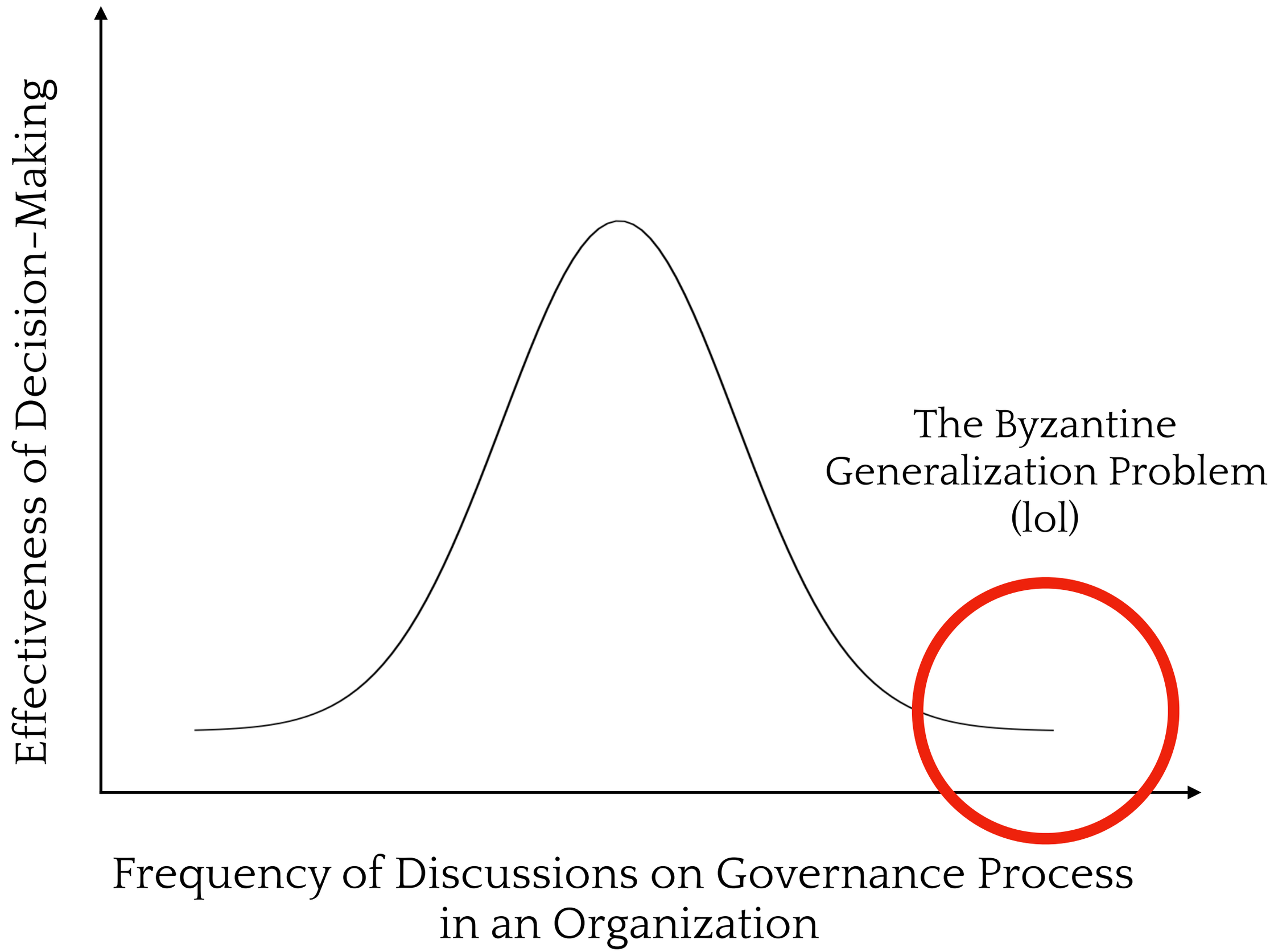
Nick Szabo ⚡ @NickSzabo4 · Feb 13

It's too hopelessly broad of a term, it can be made to mean almost anything. Its use doesn't shed any light on the actual residual human decision-making that is needed.

Argument Surface




Effectiveness of Decision-Making
Frequency of Discussions on Governance Process
in an Organization


Effectiveness of Decision-Making
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(lol)
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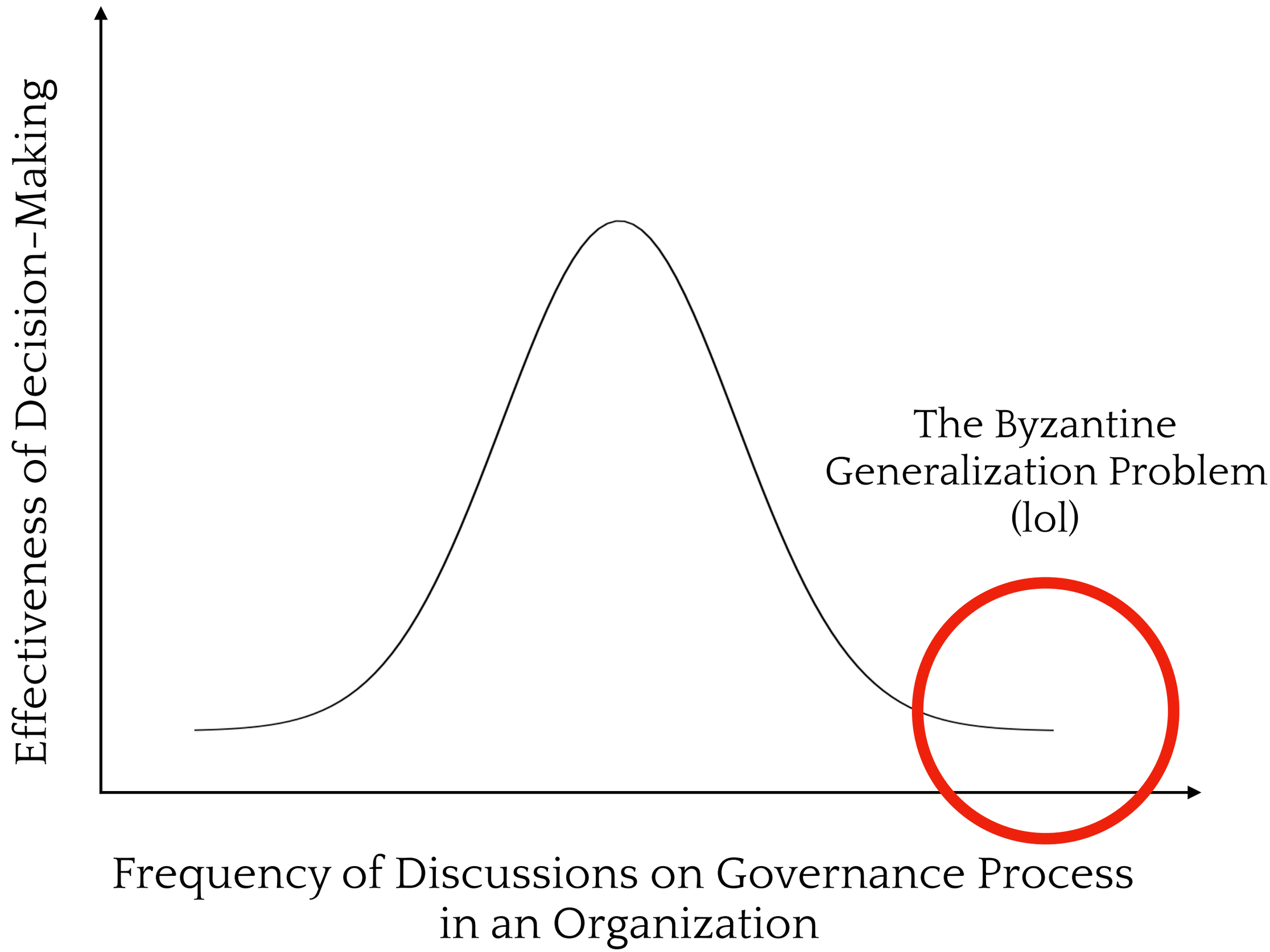
SegWit 2x

Our goal has always been a smooth upgrade for Bitcoin. Although we strongly believe in the need for a larger blocksize, there is something we believe is even more important: keeping the community together. Unfortunately, it is clear that we have not built sufficient consensus for a clean blocksize upgrade at this time. Continuing on the current path could divide the community and be a setback to Bitcoin's growth. This was never the goal of Segwit2x.

Ethereum Hard Fork

In order to reach a quick consensus, the hard fork proposal was voted on and approved by Ether holders, who had to send a transaction to a voting platform. The super majority of people (89%) voted for the Hard-Fork and it took place during the 1920000th block (20th July 2016).

Stop the DAO
You can't stop the DAO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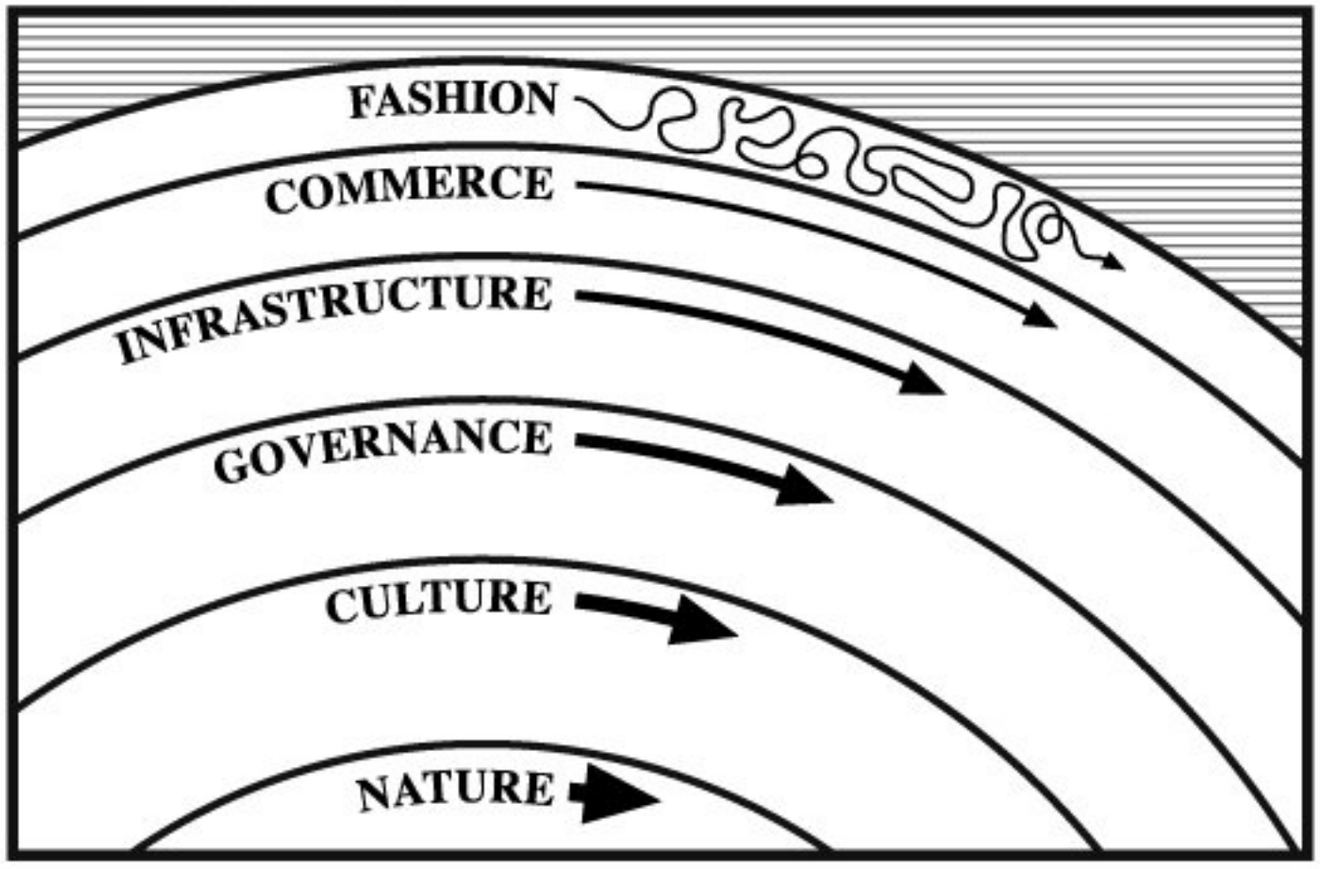
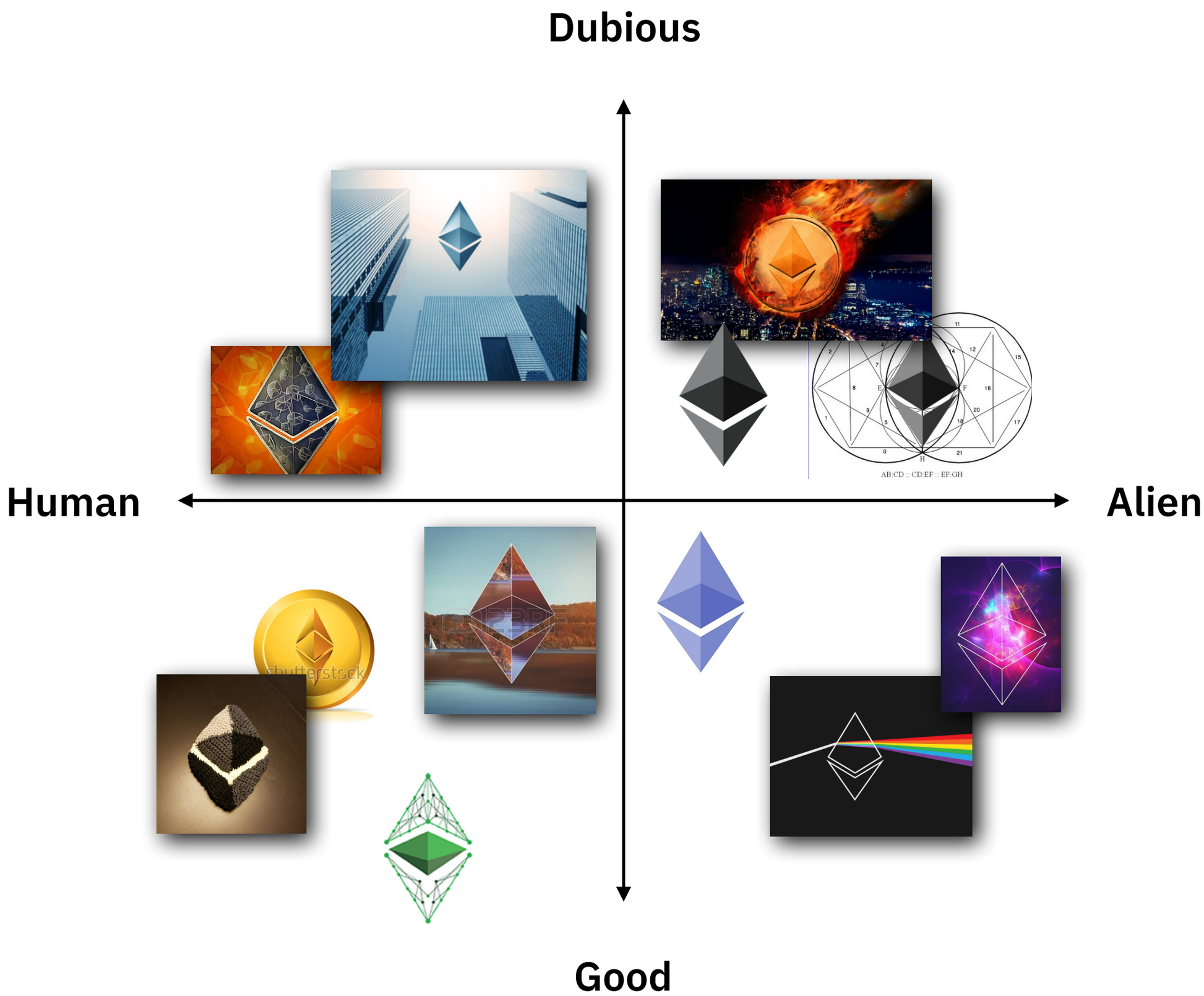
Pace Layering



Source: Brand, S., 1999, *The Clock of the Long Now*, p. 37.


Dubious
Human
Alien
Good
AB:CD :: CD:EF :: EF:GH

The Byzantine Generalization Problem

A strategy to be used sparingly